AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT395
Applied Cybersecurity

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 20/Nov./2025

Number of Exam Pages: 3	Time Allowed: 90 m

Instructions:

- Total Marks: 30
- Total Number of Questions:3
- Marks Distribution:
  - Question 1: 10 marks
  - Question 2: 10 marks
  - Question 3: 10 marks
- Write your answer in the answer booklets. Answers given on the exam paper will not be marked.
- The use of electronic devices that could have a memory is NOT permitted
- Be sure you write your name and ID on the External Answer booklet.

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Question 1: Select the best answer : **Total: 10 Marks**

1. **What type of malware disguises itself as legitimate software to trick users into installing it?**
   a) Virus
   b) Worm
   c) Trojan
   d) Ransomware
2. **Which of the following is an example of a social engineering attack?**
   a) SQL Injection
   b) Phishing
   c) DDoS Attack
   d) Brute Force Attack
3. **What is the main purpose of a Distributed Denial-of-Service (DDoS) attack?**
   a) To steal confidential information
   b) To render a system unusable by overwhelming it with traffic
   c) To exploit vulnerabilities in an application
   d) To access a network without authorization
4. **Which network attack involves manipulating the Domain Name System (DNS) to redirect traffic to malicious websites?**
   a) DNS Poisoning
   b) ARP Spoofing
   c) SQL Injection
   d) Session Hijacking
5. **Which wireless security protocol is considered the weakest and should not be used?**
   a) WPA2
   b) WPA3
   c) WEP
   d) WPA
6. **Which method provides an extra layer of protection by encrypting individual files or folders?**
   a) Full Disk Encryption (FDE)
   b) File-level encryption
   c) BitLocker
   d) Hardware Security Module (HSM)
7. **Which of the following is a proactive approach to finding security weaknesses in a system?**
   a) Vulnerability scanning
   b) Incident response
   c) Encryption
   d) Firewalls
8. **Which of the following is an example of a threat actor?**
   a) Antivirus software
   b) Security policy
   c) Hacker
   d) Firewall
9. **Which type of attack specifically targets a specific individual or organization using personalized tactics?**
   a) Whaling
   b) Smishing
   c) Vishing
   d) Pharming
10. **What is a buffer overflow attack?**
    a) An attack where an attacker floods a network with traffic
    b) An attack that forces too much data into a memory buffer, causing it to overwrite adjacent memory
    c) An attack where multiple systems are used to target a single system
    d) An attack that intercepts network communications

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Question 2: Describe the following general kinds of attacks, in a few sentences: Total: 10 Marks

1. How does a brute force attack work, and what makes it a threat to network security?
2. What is the role of a proxy server in enhancing network security?
3. How does a phishing attack exploit human behavior, and what are some common delivery methods?
4. What are the security benefits and challenges of using VLANs (Virtual Local Area Networks)?
5. How does a Distributed Denial of Service (DDoS) attack differ from a standard DoS attack?

Question 3 Answer the following questions [10 Marks]

1. What is the purpose of encryption in cybersecurity? (2 marks)
2. What are the primary characteristics used to classify social engineering attacks? (2 marks)
3. What are packet sniffing attacks, and how can they be mitigated? (2 marks)
4. What is the difference between a virus and a worm in terms of propagation? (2 marks)
5. What is the role of a security policy in administering a secure network? (2 marks)

End of questions

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